Behavioural Traces as Neural Artifacts: An Exact Null for Model Populations, and Lineage Recovery Without Weights

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Abstract

Learning from populations of neural artifacts has so far meant learning from weights, gradients and internal traces. For most of the open model ecosystem none of those are practically available at population scale, but one artifact always is: the record of which benchmark items a model gets wrong. We argue this behavioural trace deserves treatment as an artifact modality in its own right, and we supply the three things such a modality needs. First, a released model zoo: per-item outcome matrices for $1{,}228$–$4{,}562$ open models across five benchmarks, harvested from archived evaluation files at zero inference cost, with the two identifier pitfalls that make a naive harvest return an empty join documented and guarded. Second, an exact null. We show the obvious population-level statistic, mean pairwise co-failure, is a deterministic function of item difficulty alone and therefore carries no information about model relatedness whatsoever, and that the row and column margins of the outcome matrix are the jointly sufficient statistics of the Rasch family, so the uniform distribution over margin-matched matrices is a fit-free conditional null that removes item difficulty exactly. Third, evidence the modality carries provenance: after conditioning, a 5-nearest-neighbour classifier on the leading residual eigenvectors recovers a model's base family with leave-one-out accuracy 0.807 over 916 labelled models from five families, against a label-permutation null of 0.287 ± 0.017 and a majority-class rate of 0.382, while the same pipeline run on a draw from the exact null returns 0.318, at chance. We report the controls that weaken the strongest reading as prominently as the result, including that unconditioned eigenvectors already reach 0.760, so conditioning is not what creates the signal.

1 Introduction

Model repositories now hold hundreds of thousands of trained networks, and a growing body of work treats that population as data: learning on weights, merging models, predicting properties from parameters, mapping model atlases. The premise is that a trained model leaves artifacts, and that populations of artifacts can be compared, searched and explained.

That premise has an access problem which the weight-centric framing does not solve. Weights are downloadable in principle and prohibitive in practice at population scale; gradients, optimization trajectories and internal representations require running the model, which for thousands of models is a serious compute bill; and for any model behind an API, none of the four exists. What does exist, for every model anyone has ever evaluated, is the record of which items it got wrong. Public leaderboards have been accumulating exactly this for years, item by item, model by model, and it is readable without loading a single weight.

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We take that record seriously as an artifact modality. A model's per-item outcome vector is a computational trace: it is a projection of the weights through a fixed probe set, lossy but faithful, and cheap enough that a population of thousands is a file read rather than a compute job. The question this paper asks is whether that trace supports the things this community wants from artifacts, and specifically whether it carries *neural lineage*.

Contributions.

1. **A model zoo in the behavioural-trace modality** (Section 3): per-item binary outcome matrices for five benchmarks over 1,228–4,562 models each, with the harvest procedure, the two identifier pitfalls that silently break it, and the row-order audit that licenses the join.
2. **An exact, fit-free null for population-level analysis of these artifacts** (Section 4). We prove the first statistic anyone computes is degenerate, identify the canonical null from Rasch sufficiency, and validate the sampler against exhaustive enumeration rather than asserting it.
3. **A calibrated measurement of residual structure** across the five benchmarks (Section 5), together with a negative result about the statistic most tempting to report.
4. **Lineage recovery from behavioural traces alone** (Section 7): family classification at 0.807 against a 0.287 permutation null, with five pre-registered controls, including two that cut against the strongest reading.
5. **A monitor for a growing population** (Section 8), since a model zoo harvested from a live archive is not a fixed dataset.

2 Related work

Learning from populations of neural artifacts has largely meant learning from weights. Unterthiner et al. (2021) showed a trained network's accuracy is predictable from simple weight statistics without evaluating it on data, and released 120k CNNs to support the task; Schürholt et al. (2022) published systematically generated model zoos as a dataset in their own right. Our zoo is complementary in construction and in cost: it is not generated but harvested, the models are the ecosystem's actual submissions rather than a controlled sweep, and the artifact is a behavioural trace, which is available for models whose weights are impractical to collect or absent entirely. The trade is real in both directions — weights support parameter-space operations such as merging and editing that a trace cannot, while a trace is obtainable at population scale for essentially free and is architecture-agnostic by construction.

Closest to our lineage result is the model atlas of Horwitz et al. (2025), who chart Hugging Face's documented lineage graph, demonstrate applications including attribute prediction, and, because "most models are poorly documented", propose recovering undocumented regions from structural priors on real-world training practice. We attack the same documentation gap from the opposite side. We quantify it on our population — a declared parent resolves for 23.3% of models, with 315 of 1,362 repositories no longer retrievable — and then ask what can be recovered with no documentation and no priors on training practice at all, using only behaviour. The two are complementary: structural priors and behavioural evidence are independent sources on the same latent graph, and a signal that recovers family at 0.807 from traces is a candidate edge weight for exactly the undocumented regions an atlas cannot yet chart.

Methodologically, the exact conditional null we use is standard in ecology (Schluter, 1984; Gotelli, 2000; Strona et al., 2014) and psychometrics (Rasch, 1960; Andersen, 1973; Fleiss, 1971), and the degeneracy of the mean statistic is known independently in both plus the classifier-ensemble literature (Kuncheva & Whitaker, 2003; Cronbach, 1951). None of it appears in the artifact-population literature, and importing it is a large part of what this paper is for.

3 The artifact and the zoo

Let $F \in \{0,1\}^{N\times M}$ with $F_{im} = 1$ iff model i fails item m; row sums r_i, column sums c_m. One row is one model's behavioural trace on one probe set.

Benchmark	frozen analysis set models	items	current release models	items	mean accuracy
ARC-Challenge	1362	1165	4562	1170	0.527
Winogrande	1361	1267	4544	1267	0.734
TruthfulQA (mc1)	1334	786	3531	811	0.368
GSM8K	1228	1319	3405	1319	0.348
HellaSwag	1362	9404	2972	9802	0.581

Table 1: The zoo. Counts are after removing rows and columns that are all-correct or all-wrong; mean accuracy is over the full pre-filter matrix. The archive is live and a scheduled process keeps harvesting it, so the two column groups differ: every analysis below names the snapshot it used and the release pins each by content hash.

We harvest F from the archived Open LLM Leaderboard. Only the metric column of each result file is read — $1.4\,\text{KB}$ out of a $63\,\text{MB}$ file for HellaSwag — which is what makes an ecosystem-scale harvest possible at zero inference cost.

Two pitfalls, documented because they are silent. Item identifiers are not stable across the archive: the harness stored a dataset identifier in mid-2023 and the raw question text thereafter, so a naive identifier join returns an *empty* intersection rather than a wrong one. The accuracy field also moved into a nested `metrics` struct in the 2024 schema. We resolve item identity by row index, licensed by explicit verification rather than assumption: across all five benchmarks, row order was identical for every one of 334 models sampled evenly over July 2023 to May 2024, with zero mismatches (ARC $149/149$, TruthfulQA $60/60$, GSM8K $60/60$, Winogrande $40/40$, HellaSwag $25/25$), and a row-count guard runs on every read. An undetected misalignment would make a model's row look independent of every other and so bias correlations *toward* the null, understating rather than manufacturing any result below.

The zoo is not static, and this matters. A population harvested from a live archive grows underneath the analysis. Between the frozen set and the current release, ARC grew from $1{,}362$ to $4{,}562$ models; one harvest landed during the preparation of this paper. Every number below is therefore labelled with its snapshot, and each snapshot is identified in the released artifacts by the SHA-256 of the matrix file, which is what makes a run reproducible when the underlying archive will not hold still. We think any benchmark or dataset in this modality needs the same discipline, and note it as a small proposal for the community's dataset practice.

4 Why population-level analysis of these artifacts needs a null

The natural first question about a population of traces is how often two models fail together. That statistic is uninformative, and provably so.

Lemma 1 (Mean co-failure depends only on item margins)**.** *The mean pairwise co-failure rate is* $O = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{M} \sum_m F_{im} F_{jm} = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$.

Proof. $\sum_{i \neq j} F_{im} F_{jm} = (\sum_i F_{im})^2 - \sum_i F_{im}^2 = c_m^2 - c_m$ since $F_{im} \in \{0, 1\}$. $\square$

O is a function of the item margins alone, so any randomization that preserves them leaves it invariant with exactly zero variance: the "excess" co-failure over such a null is identically zero by construction, not empirically small. Mean pairwise disagreement inherits the degeneracy, being $2\bar{p} - 2O$. The same identity is Schluter's V-ratio in ecology (Schluter, 1984; Gotelli, 2000), Cronbach's α in psychometrics (Cronbach, 1951), and Fleiss's observed agreement in the multi-category case (Fleiss, 1971); it is also the double-fault diversity measure of the classifier-ensemble literature (Kuncheva & Whitaker, 2003). We restate rather than claim it, because a population-level artifact study that reports mean co-failure is reporting item difficulty.

The canonical null. The Rasch model $\Pr(F_{im} = 1) = \sigma(\alpha_i + \beta_m)$ has $\{r_i\}$ and $\{c_m\}$ as jointly sufficient statistics, so the conditional law of F given its margins is uniform on the set of margin-

		rms $\lvert R_{ij}\rvert$			participation ratio		
Benchmark	N	obs.	null	ratio	obs.	null	dims > edge
ARC-Challenge	1362	0.2483	0.0449	5.5×	23.6	449.3 ± 3.4	6
Winogrande	1361	0.2691	0.0409	6.6×	20.1	493.8 ± 3.0	5
TruthfulQA	1334	0.2942	0.0583	5.1×	17.4	309.7 ± 2.4	5
GSM8K	1228	0.0987	0.0346	2.9×	118.6	553.1 ± 2.2	4
HellaSwag	1362	0.2457	0.0206	11.9×	27.3	958.0 ± 3.2	12

Table 2: Residual structure on the frozen analysis set after exact-margin conditioning. Null rms is the mean over 40 curveball replicates; null PR is mean ± SD over 60 independent-chain replicates, both margins exactly preserved at every draw.

matched binary matrices, for every parameter value (Rasch, 1960; Andersen, 1973). That set is the ecologists' fixed-fixed null. Conditioning on it removes model ability and item difficulty exactly, without an optimizer, a penalty, or an identification convention — which matters here because the alternative, fitting an item-response model to each population, introduces analyst choices that a population study should not have to defend.

Sampling it, verified rather than asserted. We sample with curveball (Strona et al., 2014), whose convergence to uniform on the fibre requires that failed trades count as steps (Carstens, 2015); ours does. We test it directly: enumerating every binary matrix with given margins on four small fibres (sizes 48, 90, 204, 1,170) and drawing 60,000 thinned samples, a χ^2 test against uniformity is not rejected on any ($p = 0.23$, 0.05, 0.73, 0.15). On the real matrices, four dispersed chains give Gelman–Rubin $\hat{R} \in [0.98, 1.03]$ at the burn-in used.

Statistics. Let P be the Rasch mean field matching the observed margins (fitted to margin error below 10^{-11}), $D = (FF^\top - PP^\top)/M$, and R the unit-diagonal scaling of D. We report the rms off-diagonal $\lvert R_{ij}\rvert$, the participation ratio $\mathrm{PR} = (\sum_k \lambda_k^+)^2 / \sum_k (\lambda_k^+)^2$ of R's positive spectrum, and the count of eigenvalues of R above the null's largest — parallel analysis against the exact null rather than against iid noise. Each is compared to the same quantity on margin-matched replicates. P is validated against the randomization itself: over 319,600 ARC model pairs, the curveball expectation of $\frac{1}{M}\sum_m F_{im}F_{jm}$ and the Rasch prediction correlate at 0.99999.

5 What the null reveals

Table 2 reports what survives conditioning. Residual correlation runs 2.9–11.9× its null and the participation ratio sits at 2.8–21.4% of its null on every benchmark. Before conditioning, the same participation ratio reads 1.5–3.6 on all five, a number driven almost entirely by the single shared item-difficulty eigenvalue that conditioning removes: uncalibrated, the modality looks like it contains almost no independent models, which is an artifact of the statistic rather than a property of the population.

A negative result about the tempting statistic. It is tempting to read Table 2 as "1,362 models behave like 24". That reading is wrong for an algebraic reason worth stating in this venue, because the participation ratio is a natural summary for any artifact population. For the unclipped spectrum, $\mathrm{PR} = N/(1 + (N-1)\overline{R_{ij}^2})$ exactly — 16.04 on ARC computed either way — so PR is a monotone transform of mean squared residual correlation and the eigendecomposition contributes nothing. An equicorrelated matrix and a partition into 24 tight clusters give the same value. The statistic cannot license a count of independent models, and we report it only as a calibrated summary. The diagnostics that *can* separate those cases say the ARC structure is neither: the leading eigenvalue carries 54% of squared residual mass (one shared factor would give ≈ 99%, twenty clone blocks ≈ 10%), deflating it raises PR only to 46.8 rather than toward N, and six eigenvalues clear the null's spectral edge.

What else could produce this. Each row below is a simulated population with no clusters, no clones and no shared lineage — errors conditionally independent given the item.

Generating process	PR / PR_{null}	rms $\lvert R_{ij} \rvert$	$\lambda_1^2/\sum\lambda^2$	dims > edge
1PL, correctly specified	1.001	0.065	0.105	1
2PL, $\text{sd}(\log a) = 0.35$	0.966	0.068	0.119	1
2PL, $\text{sd}(\log a) = 0.60$	0.862	0.076	0.167	1
two ability dimensions	0.230	0.162	0.457	3
20 exact clone clusters	0.112	0.234	0.102	19
real ARC	**0.052**	**0.248**	**0.540**	**6**

Table 3: Discrimination heterogeneity alone cannot reproduce the observed effect. A low-dimensional multi-ability structure reproduces much of it, and the observed spectral signature resembles that case rather than duplicated models.

Two readings follow, and the second is a limitation. Discrimination heterogeneity is not a sufficient explanation: even at $\text{sd}(\log a) = 0.60$ it yields rms 0.076 against the observed 0.248. But a small number of extra ability dimensions is a substantial one, and the observed $\lambda_1^2/\sum\lambda^2 = 0.540$ sits near the two-dimensional case (0.457) and far from the clone case (0.102). We therefore cannot separate "genuine behavioural concentration" from "the conditioning family is too small", and on the spectral evidence the latter is the better description.

Robustness. Recomputing every statistic on populations deduplicated at pairwise agreement 0.95, discarding 21–64% of models with the Rasch fit and null redrawn each time, moves 14 of 15 statistic–benchmark pairs by at most $1.5\times$; the exception is HellaSwag's PR ratio at $2.44\times$. On planted data the test reaches power 1.00 against shared failure modes at a planted rms of 0.048, far below the 0.099–0.294 observed, and rejects at 0.017 under the null. Against the alternative in which every model shares one item-difficulty profile and differs only in ability, power is exactly α — that alternative *is* the null, and we report the blind spot as a property of the design rather than a caveat. Two candid points about the null itself: its rms sits only 1.25–$2.01\times$ above the analytic $1/\sqrt{M}$ floor, and conditioning on item margins alone reproduces the both-margin null to within 8.3% on every benchmark, so the item margins do essentially all of the conditioning work for this statistic.

6 A second trace modality: which wrong answer

Binary correctness is the coarsest reading of a behavioural trace. The archive also records per-choice log-likelihoods for part of the population, from which the model's *chosen* answer is recoverable, giving a multi-category trace: not whether a model failed, but how. Analysts use it to ask whether models make the same mistakes. That statistic has its own degeneracy, and the same conditioning idea repairs it.

Lemma 2 (Agreement given both wrong depends only on per-item response composition)**.** *Let n_{mk} be the number of models choosing option k on item m and $n_m = \sum_k n_{mk}$. The mean pairwise same-answer rate is $A = \sum_m \sum_k n_{mk}(n_{mk} - 1)/\sum_m n_m(n_m - 1)$, a function of the per-item composition $\{n_{mk}\}$ alone. Any relabelling of which model gave which recorded answer, item by item, leaves A exactly invariant with zero variance.*

This is Fleiss's observed agreement (Fleiss, 1971) and the founding premise of answer-copying detection in psychometrics. It matters here because it is easy to demonstrate constructively: in a simulation where models err *conditionally independently* given the item, drawing from a shared per-item distractor-attractiveness distribution, agreement-given-both-wrong is 0.5567 against a uniform baseline of 0.3333 — an apparent excess of $+0.2233$ that looks like shared behaviour and is not — while the excess over the composition-preserving null implied by Lemma 2 is -1.1×10^{-16}. Independent models, large apparent error correlation, purely from the distractor structure of the probe set.

The critique does not always bite, and we report where it does not. A degeneracy result invites the conclusion that findings in this modality are artifacts. We tested that against a specific published claim — that more accurate models give more similar wrong answers — with a kill condition fixed in advance: if the accuracy–agreement slope survives conditioning with a confidence interval excluding zero, the critique does not apply and the claim stands. We reconstructed each model's chosen answer from per-choice log-likelihoods on a 600-model ARC subsample, recovering the gold answer from cross-model agreement (agreement among correct models 1.0000 on 1,144 items; the recovered key reproduces each model's own reported accuracy on 99.01% of cells, which we take as validating the reconstruction rather than assuming it).

Over 38,238 model pairs, the raw slope of agreement-when-both-wrong against mean pair accuracy is $+0.585$ (95% CI $[0.577, 0.594]$); conditioned on the composition-preserving null it is $+0.514$ ($[0.506, 0.522]$), an attenuation of only 12.1%. The kill condition fired: the claim is not an artifact of item-level distractor composition and we report it as robust. Two things are true at once, and both are properties of the modality worth knowing. The mean-level excess is essentially zero — 0.7332 observed against 0.7324 expected under conditional independence, $+0.00085$, exactly as Lemma 2 predicts — while the *relationship* between accuracy and agreement is not a mean-level statistic and is not absorbed by the same conditioning. Degeneracy of a summary does not imply degeneracy of every claim built on the underlying trace, and the way to tell is to condition and look.

Two opposite-looking results reconcile exactly. A related trap: on the binary traces, a pre-registered dispersion statistic $\mathrm{Var}_{i<j}(C_{ij})$ came out *below* its null on three of five benchmarks, while the participation ratio sits far below its null, which is algebraically the same as residual correlations sitting far *above* theirs. Both are correct. Writing $C = E + D$ with $E = PP^\top/M$ the margin-model part, $\mathrm{Var}(C) = \mathrm{Var}(E) + 2\,\mathrm{Cov}(E, D) + \mathrm{Var}(D)$, and because the null preserves margins, $\mathrm{Var}(E)$ is identical for the observation and every replicate. On ARC, $\mathrm{Var}(D) = 6.25 \times 10^{-4}$ against a null of 2.13×10^{-5} (a $29\times$ excess), while $\mathrm{Cov}(E, D) = -6.49 \times 10^{-4}$ against -2.1×10^{-6} (correlation -0.285 against -0.005). The negative covariance outweighs the inflated variance, which is exactly why the dispersion reads low. Pairs the margin model expects to co-fail most actually co-fail relatively less: structured deviation from unidimensionality, which is what "models cluster into behavioural types" means when stated in these terms.

7 Neural lineage from behavioural traces

The previous section says the modality contains structure. This section asks whether that structure is *about* anything, and takes the question this workshop poses directly: can artifacts reveal provenance?

Why we could not answer it with metadata. We first tried the direct route. The leaderboard carries an optional, self-reported `base_model` field; on the frozen ARC set it resolved a declared parent for 317 of 1,362 models (23.3%), with 315 repositories no longer retrievable at all. A 40% coverage threshold had been registered in advance as the point below which a lineage decomposition would not be trusted. It fired, and the decomposition was dropped rather than run on the surviving quarter. Declared provenance for this population does not exist at usable coverage, which is precisely the condition under which an artifact-derived signal would be worth having.

Setup. On the pinned ARC snapshot ($N = 3{,}762$, $M = 1{,}169$, SHA-256 `79d2490e...`), take the top-k eigenvectors of R with k the number of eigenvalues above the exact null's spectral edge ($k = 4$, edge 79.5). Label base family by string matching on the model identifier against known base-model names — a coarse lower bound on relatedness, explicitly not lineage ground truth, reaching 34.5–35.7% of identifiers. Use the five largest families: 916 models, Mistral 350, Llama-2 211, Llama-3 188, Qwen 84, Mixtral 83. Classify by leave-one-out 5-nearest-neighbour in the standardised loading space.

Result. Family is recovered at accuracy $\mathbf{0.807}$, against a label-permutation null of 0.287 ± 0.017 (1,000 permutations, $p = 0.001$, the resolution floor) and a majority-class rate of 0.382. Per-family recall is above chance for four of five: Mistral 0.926, Llama-2 0.839, Llama-3 0.777, Qwen 0.714, Mixtral 0.386. Accuracy rises monotonically with the number of eigenvectors used — 0.452 at $k = 1$, 0.807 at $k = 4$, 0.864 at $k = 6$, 0.905 at $k = 20$ — so the lineage signal is spread across many residual dimensions rather than carried by one. Every control below was specified before the experiment ran.

What cuts against the strongest reading. Three things in Table 4, and we state them rather than bury them. First, margin conditioning is *not* what produces the signal: unconditioned correlation

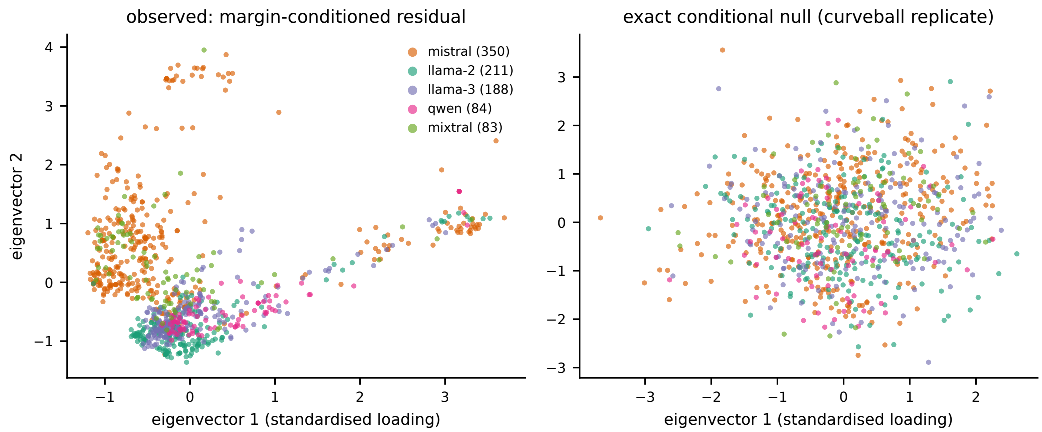

Figure 1: Models projected on the top two residual eigenvectors, coloured by base family (left), and the identical pipeline applied to a curveball replicate of the same matrix (right). The replicate preserves both margins exactly and destroys any real association between behaviour and family, so the right panel is what the left would look like if the construction were manufacturing structure. The classifier uses four eigenvectors, not the two drawn here.

Arm	What it tests	Acc.	Perm.	Reading
Main	residual eigenvectors, real labels	**0.807**	0.287	signal present
Exact-null draw	pipeline on a curveball replicate	0.318	0.289	at chance
Accuracy only	model accuracy as the sole feature	0.432	0.288	carries some, not most
Accuracy proj. out	loadings orthogonalised to accuracy	0.809	0.286	not an accuracy artifact
Raw correlation	unconditioned eigenvectors	0.760	0.287	conditioning not the source
Dedup 0.95	$N = 2{,}350$, 591 labelled	0.787	0.276	survives
Dedup 0.90	$N = 1{,}288$, 301 labelled	0.551	0.264	weakened, above chance

Table 4: Pre-registered controls. The exact-null arm is the calibration check: a pipeline reporting signal on a margin-matched replicate would be reporting an artifact of the construction. It does not.

eigenvectors already reach 0.760, so conditioning adds about five points. The exact null earns its place as the calibration arm and as what makes the mean-level statistic provably uninformative (Lemma 1), not as the source of this result. Second, part of the signal is near-duplicate structure: accuracy falls to 0.551 under deduplication at agreement 0.90, which discards two thirds of the population. The residue is still roughly twice the permutation null, so it is not only duplicates, but it is partly duplicates. Third, the labels are name strings; a model named for a base it was not derived from is mislabelled here and we cannot audit that.

What this does and does not establish. It is not an attribution method: it returns a coordinate in a residual eigenspace, not a claim about which examples caused which behaviour. An alternative explanation we cannot exclude is that the residual dimensions track training recipe or data mixture, which correlates with family without ancestry — that would make this a signature of convergent practice rather than of lineage, and separating the two needs ground truth this population lacks. What it does establish is the premise the lineage agenda needs in this modality: behavioural traces are not provenance-blind. Under a null that removes item difficulty by construction, and after projecting out model accuracy, behaviour alone locates a model in its family most of the time, at a cost of one Rasch fit and one eigendecomposition, with no access to weights, training data, or inference.

8 Monitoring a population that keeps growing

A zoo harvested from a live archive raises a question static datasets do not: has the population changed? Because the conditional null is exact at every population size, a Monte Carlo p-value for rms $|R_{ij}|$ is valid at each look, and repeated looks can be combined without the type-I inflation that repeated testing on a growing sample would otherwise incur.

We order the ARC models by submission date and take twelve monthly looks as the population grows from 241 to $3{,}762$. At each look we compute a Monte Carlo p-value against $R = 40$ curveball replicates drawn for that population, calibrate it to an e-value by $e_t = \kappa p_t^{\kappa-1}$ with $\kappa = 1/2$, and merge by the arithmetic mean, which is valid under arbitrary dependence (Vovk & Wang, 2021) — necessary here, since nested populations are as dependent as looks get. As a control we run the identical pipeline with the observed matrix at each look replaced by a draw from that look's own null.

The control is the informative arm: its per-look p-values range over $[0.049, 0.951]$ and its merged e-value ends at 0.90, never exceeding 1.17, so the procedure does not accumulate evidence when there is none. On real data every look returns the smallest p-value 40 replicates can express, so the merged e-value pins at 3.20; that is the calibrator's value at $p = 1/41$ and is a statement about Monte Carlo resolution, not a likelihood ratio. The substantive reading is the ratio itself, flat to slightly declining as the population quadruples ($5.90\times$ at $N = 241$, $5.38\times$ at $N = 3{,}762$): residual concentration in this zoo is not visibly worsening over the window, and this is the instrument that would show it if it were.

Trends across the population. The same machinery reads by release cohort rather than cumulatively. Partitioning ARC's frozen set into the eight monthly cohorts with enough models to condition, the calibrated participation ratio per cohort stays in a narrow band (14.2 to 19.9) while cohort size varies from 70 to 242, so PR/N falls from 0.230 in 2023-07 to 0.060 in 2024-03 and recovers to 0.120 by 2024-05. Read carefully, that is mostly a statement about how PR/N behaves as N grows rather than a discovered trend, which is why we report the cohort series and not a trend test: the pre-registered cohort-trend hypothesis was never confirmatorily tested and we do not claim it here.

The construction is valid for a pre-specified set of looks and not for optional stopping, which would require a null for the increment when a row arrives; conditioning does not supply one, since given the earlier rows and the new margins the arriving row is determined.

9 Limitations

The traces are correctness and chosen-answer on multiple-choice-style probes, which is a coarse projection of behaviour; the generative traces that exist for part of the archive are not exploited here, and the multi-category analysis of Section 6 covers one benchmark and a 600-model subsample rather than the full zoo. The population is self-selected leaderboard submissions, not a random sample of the ecosystem, and a different peer set can change population-level inferences. The conditioning family is unidimensional; Table 3 rules out discrimination heterogeneity and duplicate models as sufficient explanations but cannot separate genuine concentration from an underfit null. Lineage labels are name strings on one benchmark with one labelling scheme, and the five families used are the five largest. We report classification accuracy, not a calibrated provenance probability, and offer no decision procedure. Finally, all five benchmarks come from a single archive and a single harness generation; a directional check on an independently implemented successor harness (ARC, 212 models) reproduces the sign but not the magnitude ($4.1\times$ rms excess against $5.5\times$), which we report as evidence against a pure harness artifact and not as a second measurement.

10 Reproducibility

Every number above traces to a JSON artifact emitted by an executed run, and the propositions are asserted as unit tests. The study was pre-registered before the confirmatory analysis, with dated amendments recorded before any confirmatory statistic was computed; one pre-registered hypothesis was refuted by its own kill condition and is reported as refuted, and the lineage gate described in Section 7 fired and removed an analysis. Because the archive is live, each analysis names its snapshot and the release pins each snapshot by SHA-256, so a run is recoverable even after the underlying files have grown. Code, the harmonised substrate with a datasheet and Croissant metadata, and every results artifact are archived; links are withheld here for anonymity and will be restored in the camera-ready. A large language model was used as a coding and drafting assistant; the authors specified and pre-registered the hypotheses and kill conditions, verified every derivation and every reported number against an executed-run artifact, and are responsible for the content.

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